Module 07: The Accounting Equation

Course Learning Outcomes:

1. Define the Accounting Equation;
2. Enumerate and Explain the Elements of the Accounting Equations;
3. Solve Basic Problems Applying the Accounting Equation;

Introduction

Accounting equation describes that the total value of assets of a business is always equal to its liabilities plus owner's equity. This equation is the foundation of modern double entry system of accounting being used by small proprietors to large multinational corporations. Other names used for accounting equation are *balance sheet equation* and *fundamental or basic accounting equation.*

The Accounting Equation

We know that every business owns some properties known as assets. The claims to the assets owned by a business entity are primarily divided into two types – the claims of creditors and the claims of owner of the business. In accounting, *the claims of creditors are referred to as liabilities* and *the claims of owner are referred to as owner's equity.*

Accounting equation is simply an expression of the relationship among assets, liabilities and owner's equity in a business. The general form of this equation is given below:

ASSETS = LIABILITIES + OWNER'S EQUITY

Notice that the left hand side (***also known as assets side***) of the equation shows the resources owned by the business and the right hand side (***also known as equity side***) shows the sources of funds used to acquire the resources. All assets owned by a business are acquired with the funds supplied either by creditors or by owner. In other words, we can say that the value of assets in a business is always equal to the sum of the value of liabilities and owner's equity.

Elements of the Accounting Equation

Assets

Assets are monetary or economic resources which are owned by an entity and are expressed to benefit it in future. Moreover they must be quantified and expressed in monetary (dollar) terms.

Some items like company's outstanding reputation, customers' loyalty, its popular brands and its killed and experienced work force etc., though benefit the business but cannot be quantified and expressed in monetary terms. In the absence of any objective monetary value these items are not reported as assets in the accounting records.

Examples of Assets :

- ***Cash -*** Generally it is the money that we use comprising of the bills and coins we use in our everyday lives in order to buy the goods that we want ab also avail the services that we need. However, when accounting for cash, we also consider cash as money that is deposited in the banks and undeposited checks from customers.
- ***Accounts Receivable -*** This represents amounts that are collectible from customers. They arise when business sells its good or services on account or on credit.
- ***Inventories -*** Per PAS No. 2, these assets which are (a) held for sale in the ordinary course of business; (b) in the process of production for such sale; (c) in the from of materials or supplies to be consumed in the production process or in the rendering of services.
- ***Equipment -*** Per PAS No. 16, these are intangible assets that are held by an enterprise for use in production or supply of goods or services, or for rental to others, or for administrative purposes and which are expected to be use during more than one period.
- ***Land and Building -*** In most business a physical store is necessary for them to operate.

Liabilities

Liabilities are obligations of an organization to pay to other entities including individuals, government and financial institutions, or other business. They represent amounts owed to lenders and suppliers.

Examples of Liabilities:

- ***Accounts Payable -*** Are funds you owe others—they sent you an invoice that is still "payable" by you.
- ***Unearned Revenue -*** is any revenue that you collect from your customers before earning it—an up-front deposit on a big web design project, a booking fee for a stay at your bed and breakfast, or a retainer for legal services.

Owner's Equity

The equity of a business is defined as the value of the assets minus the value of the liabilities. In other words the equity is the financial value that would be left if all the assets were sold and the money from the sale was used to pay off all the liabilities. Another way of

expressing this is to say that the equity is the amount of money that would be released if the business was to be wound u

Examples of Owner's Equity:

- ***Revenues-*** A business earned revenue when it sells its products or services. Other term used in revenue are sales, rent, fees etc.
- ***Expenses -*** are costs of doing the business as all businesses have to make payments for a number of benefits and services they receive. In other words, expenses are incurred when the assets of a business are used to acquire goods and services which are needed by a business to produce revenue. They do not have a lasting benefit for the business.
- ***Capital-*** The capital account of the equity represents the net investment of the business. This means that any contribution of the owner increases the assets of the business or decreases liabilities will increase the capital account.

Using the Accounting Equation

When using accounting equation, one must remember the accounting identity which means that the equality must be maintained throughout all transactions. The reason for this is because the assets of the entity will always be claimed by another party.

Also, another way to explain this is because of the identity property of mathematics which states that both side of an equation should be equal to each other at all times.

It is a golden rule that ***"Accounting equation remains balanced all the time"***. This is because of the reason that any change resulting from the business transaction also balances its equation simultaneously. Business transaction may affect either only one element (Assets, Liabilities or Capital) or two elements, out of the three elements simultaneously (i.e., Assets & Liabilities, Assets & Capital or Liabilities & Capital). In some cases, business transactions may affect all the three elements, simultaneously (i.e., Assets, Liabilities and Capital) in a single transaction.

Recall that the accounting equation as follows:

ASSETS = LIABILITIES + OWNER'S EQUITY

Let us also note that each elements of the equation has different elements involved. For example, assets includes cash, accounts receivable, inventories, equipment etc. Thus, we can now expand the equation by a basic break down of its components.

ASSETS			+	LIABILITIES		+	OWNER'S EQUITY		
Cash	Accounts Receivable Inventories	Equipment Land and Building Intangible Assets		Accounts Payable	Unearned Revenue		Owner's Capital	Revenue	Expenses

Given these definitions, let us have some examples in order to visualized its proper use.

Transaction 1: ***Initial Investment***

Mr. Rainbow starts off with a trading business by putting his $50,000 savings in the business bank account.

ASSETS	+	LIABILITIES	+	OWNER'S EQUITY
Cash $50,000				Owner's Capital $50,000
$50,000	=	$50,000		

Transaction 2: ***Purchasing of Furniture***

Mr. Rainbow got an opportunity to have a financing from ABC bank for purchase of office furniture costing $10,000.

ASSETS		+	LIABILITIES	+	OWNER'S EQUITY
Cash $50,000 (-$10,000)	Furniture $10,000				Owner's Capital $50,000
$40,000	$10,000				$50,000
$50,000		=	$50,000		

Transaction 3: ***Purchase of Asset through credit***

Mr. Rainbow acquires a suitable business premises for $20,000 on account/credit.

ASSETS			+	LIABILITIES	+	OWNER'S EQUITY
Cash $50,000 (-$10,000)	Furniture $10,000	Premises $20,000		Accounts Payable $20,000		Owner's Capital $50,000
$40,000	$10,000	$20,000		$20,000		$50,000
$70,000			=	$70,000		

Transaction 4: ***Payment of Expenses***

Mr. Rainbow paid for the electric bill for this month, $500.

ASSETS			+	LIABILITIES	+	OWNER'S EQUITY	
Cash $50,000 (-$10,000) (-$500)	Furniture $10,000	Premises $20,000		Accounts Payable $20,000		Owner's Capital $50,000	Expenses (-$500)
$39,500	$10,000	$20,000		$20,000		$50,000	-$500
$69,500			=	$69,500			

Transaction 5: ***Sales of Product***

A customer bought $25,000 worth of product from Mr. Rainbow. Half in which using cash. The other half was to be paid in 5 days

ASSETS				+	LIABILITIES	+	OWNER'S EQUITY		
Cash $50,000 (-$10,000) (-$500) $12,500	Furniture $10,000	Premises $20,000	Accounts Receivable $12,500		Accounts Payable $20,000		Owner's Capital $50,000	Revenue $25,000	Expenses (-$500)
$52,000	$10,000	$20,000	$12,500		$20,000		$50,000	$25,000	-$500
$94,500				=	$94,500				

Concept of Dual Aspect and Accounting Equation

Based on the business transactions above, the concept of dual aspect is a matter of common observation, an everyday give-and-take phenomenon. In financial terms, it means that every transaction has two aspects. Every time a business transaction occurs, it affects the accounting equation in such a way that the equilibrium of assets, liabilities and capital remains intact. This means a change in the amount of total assets is necessarily complemented by an equal change in the total of liabilities and equity (capital).

The equilibrium of accounting equation provides the very foundations on which the whole accounting structure rests. It holds true for all situations, no matter what financial transaction take place.

References and Supplementary Materials

Books and Journals

1. Joselito G. Florendo; May 2016; Fundamentals of Accountancy, Business, and Management 1 First Edition; 856 Nicanor Reyes Sr. St., Sampaloc, Manila; Rex Book Store, Inc (RBSI)
2. Philippine Financial Reporting Standards (Adopted by SEC as of December 31, 2011)

Online Supplementary Reading Materials

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2. Accounting Theory; http://www.responsive.co.nz/theory.html; November 8, 2019

3. Business Transactions and Accounting Equations; http://www.accountingnotes.net/accounting-equation/business-transactions-and-accounting-equation/3846; November 8, 2019

4. Accounting, bookkeeping, and tax tips to help you understand your small business finances.; https://bench.co/blog/; November 8, 2019

Online Instructional Videos

1. The Accounting Equation;http://www.accounting101.org/accounting-equation/; November 8, 2019

2. The Accounting Equation;https://www.youtube.com/watch?v=3y-ZfzoER_I#action=share; November 8, 2019

3. ACCOUNTING EQUATION for BEGINNERS: Assets, Liabilities and Equity; https://www.youtube.com/watch?v=56xscQ4viWE; November 8, 2019